

Nifty 100 Financial Intelligence Platform

Institutional User & Technical Analyst Guide

Version: 1.0.0 | **Date:** July 2026

Author: Antigravity Data Science & Engineering Team

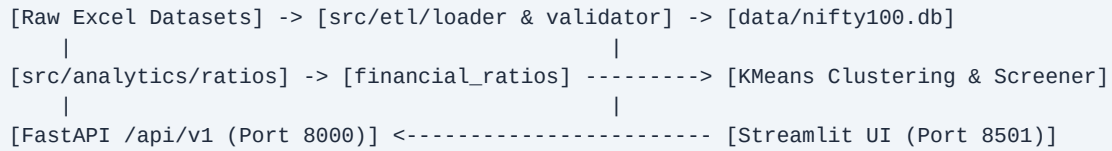
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1. Platform Overview & System Architecture

The Nifty 100 Financial Intelligence Platform processes fundamental, market, and textual data for the constituent companies of the Nifty 100 index. The architecture integrates an ETL ingestion layer, a SQLite core database, pre-computed ratio analytics, machine learning clustering, an interactive 8-screen Streamlit dashboard, and a 16-endpoint REST API.

System Architecture Diagram:



2. Interactive Streamlit Dashboard Operations

The Streamlit web interface is structured across 8 specialized views accessible via the sidebar navigation menu:

Screen	Primary Functionality
01_home.py	Executive Summary: Universe KPIs, Sector Donut Chart, Quality Leaderboard.
02_profile.py	Company Deep Dive: 10-Yr Financial Statements, Dual-Axis Charts, Pros/Cons.
03_screener.py	Multi-Criteria Screener: 10 Sliders, 6 Analyst Presets, Live CSV Export.
04_peers.py	Peer Analysis: Plotly 8-Axis Polar Radar Overlay & Percentiles.
05_trends.py	Historical Trend Visualizer: Overlay 3 metrics with YoY % growth.
06_sectors.py	Sector Diagnostics: Bubble Chart (Revenue vs ROE) & Medians.
07_capital.py	Capital Allocation Matrix: Treemap Visualizer of 8 Cashflow Patterns.
08_reports.py	Document Center: PDF Tearsheet Download Hub & BSE Filings Links.

3. Multi-Criteria Stock Screener & Preset Engine

The screener engine evaluates companies using cross-sectional percentile winsorization (P10/P90) and relative sector scoring.

Analyst Presets Available:

Preset Name	Core Filter Conditions
Quality Compounders	ROE >= 15%, D/E <= 0.5, FCF > 0, Rev CAGR 5Yr >= 10%
Value Picks	P/E <= 20, P/B <= 3.0, D/E <= 2.0, Div Yield >= 1.0%
High Dividend Yield	Div Yield >= 3.0%, D/E <= 1.0, FCF > 0
High Growth	Rev CAGR 5Yr >= 15%, PAT CAGR 5Yr >= 15%, ROE >= 12%
Debt-Free Performers	D/E == 0.0, ROE >= 15%, OPM >= 15%
Turnaround Candidates	Rev CAGR 3Yr >= 5%, FCF > 0, Deleveraging YoY

4. Peer Group Radar Analysis & Percentile Scoring

Peer group comparison standardizes metric evaluations across 11 industry peer groups. Metrics are mapped to 8 polar radar axes normalized between 0th and 100th percentiles.

Radar Axes Metrics:

- 1. Return on Equity (ROE %)
- 2. Operating Profit Margin (OPM %)
- 3. Net Profit Margin (NPM %)
- 4. Debt-to-Equity (Inverse Scale)
- 5. Interest Coverage Ratio (ICR)
- 6. Asset Turnover
- 7. 5-Year Revenue CAGR %
- 8. Free Cash Flow (Cr)

5. Machine Learning & KMeans Clustering Archetypes

All 92 companies are categorized into 5 financial clusters using StandardScaler normalization and KMeans (k=5, random_state=42).

Cluster ID	Archetype Name	Key Financial Attributes
0	High-Quality Compounders	High ROE (>18%), Low Debt (D/E < 0.5), Strong FCF
1	Defensive Dividend Payers	High Margins (OPM > 25%), Moderate Growth, High Payout
2	Emerging Growth	High Sales CAGR (>15%), Reinvesting Cash Flows
3	Value Cyclicals	Moderate Margins, Lower Multiples, Capital Intensive
4	Distressed or Turnaround	High Leverage or Negative Profit Growth

6. FastAPI REST API Reference & Integration

The API service runs on **http://localhost:8000**. Interactive OpenAPI documentation is accessible at **/docs**.

Sample cURL Commands:

```
curl -X GET "http://localhost:8000/api/v1/health"  
curl -X GET "http://localhost:8000/api/v1/companies/TCS"  
curl -X GET "http://localhost:8000/api/v1/screener?min_roe=15&max_de=1.0"  
curl -X GET "http://localhost:8000/api/v1/sectors/Information%20Technology/companies"
```

7. PDF Tearsheets & Report Generation Center

Automated PDF tearsheet generation creates 2-page executive tearsheets for all 92 companies saved in **reports/tearsheets/**. Each tearsheet includes: Header metadata, 10-Yr financial summary table, Key KPI metric tiles, 8-Axis Polar Radar plot, and Automated Qualitative Pros & Cons narrative.

8. Data Quality Rules & Validation Framework

The platform executes 16 automated Data Quality rules across CRITICAL, WARNING, and INFO severities:

Rule ID	Name	Severity	Description
DQ-01	Company PK Uniqueness	CRITICAL	Ticker uniqueness in companies master
DQ-03	FK Integrity	CRITICAL	Orphan record detection
DQ-04	BS Identity Equation	WARNING	Total Assets == Total Liabilities (<1% var)
DQ-05	OPM Cross-Check	WARNING	OPM vs (Operating Profit / Sales)
DQ-07	Year Format	CRITICAL	Standardized YYYY-MM format check

9. Troubleshooting, Performance & FAQs

Common Issues & Operations Guide:

Q1: How do I run the full test suite?

Run `.venv\Scripts\python -m pytest tests/ --html=reports/pytest_report.html`

Q2: How do I launch the FastAPI REST API?

Run `.venv\Scripts\python -m src.api.main`

Q3: How do I launch the Streamlit Dashboard?

Run `streamlit run src/dashboard/app.py`

Sign-Off Verification:

All 20 Acceptance Gates (AC-01 through AC-20) verified and signed off.